

Product Guide

NHD-600-E-TX/RX

4K60 Zero Latency Lossless Encoder and Decoder



WyreStorm

**Award-Winning Audio-Visual
Solutions Manufacturer**

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Product Overview

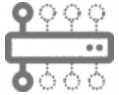


NetworkHD 600 Series Lite 4K60 Lossless Zero Latency Encoder and Decoder

The NHD-600-E encoder and decoder deliver cost-optimized, high-performance SDVoE AV over IP transmission and reception. The “E” variations remove some of the I/O found on the standard encoder and decoder, keeping only the core functionality required for most applications. Engineered for installations requiring pixel-perfect 4K60 4:4:4 video with zero-frame latency, this series supports robust 10GbE network operation with PoE+ and a flexible AV control suite.

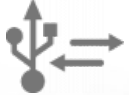
As the economic extension of the NHD 600 Series, the NHD-600-E enables integrators to deploy reliable AV distribution for point-to-point or scalable matrix applications without unnecessary complexity or cost.

Main Features



Lossless Video Quality

Deliver pixel-perfect, lossless video with support for up to 4K60 4:4:4 over HDMI 2.0, including HDR formats such as HDR10, HLG, and Dolby Vision.



Flexible Connectivity

Both the encoder and decoder feature HDMI 2.0 connectivity, along with analog audio connections and bidirectional IR and RS-232 passthrough, enabling seamless integration with external devices.



Zero-Latency Performance

Delivers zero-frame latency for real-time video transmission. This ensures perfectly synchronized audio and video, making it ideal for live presentation environments, control rooms, and interactive AV systems where timing accuracy is critical.



Cost-Effective Solution

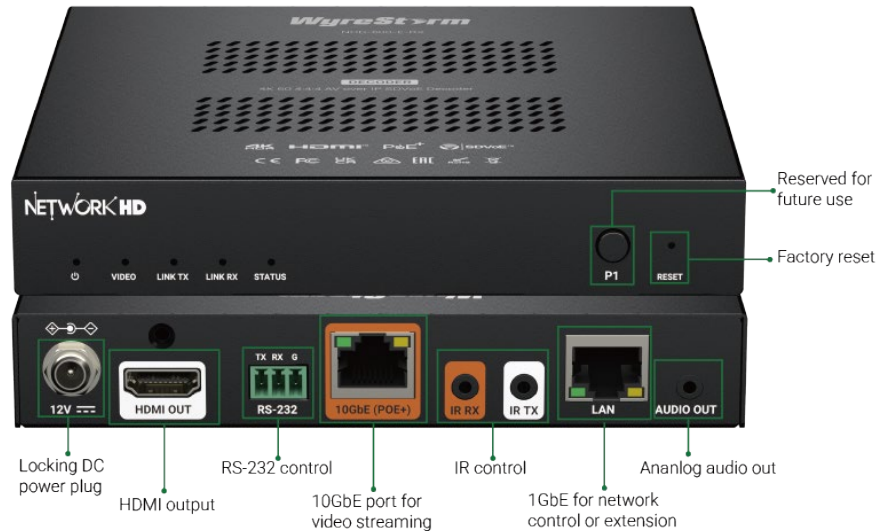
Remove some of the I/O found on the standard encoder and decoder and retain only the core functions required for most applications

Interface of the Product

NHD-600-E-TX



NHD-600-E-RX



Specifications & Packing List

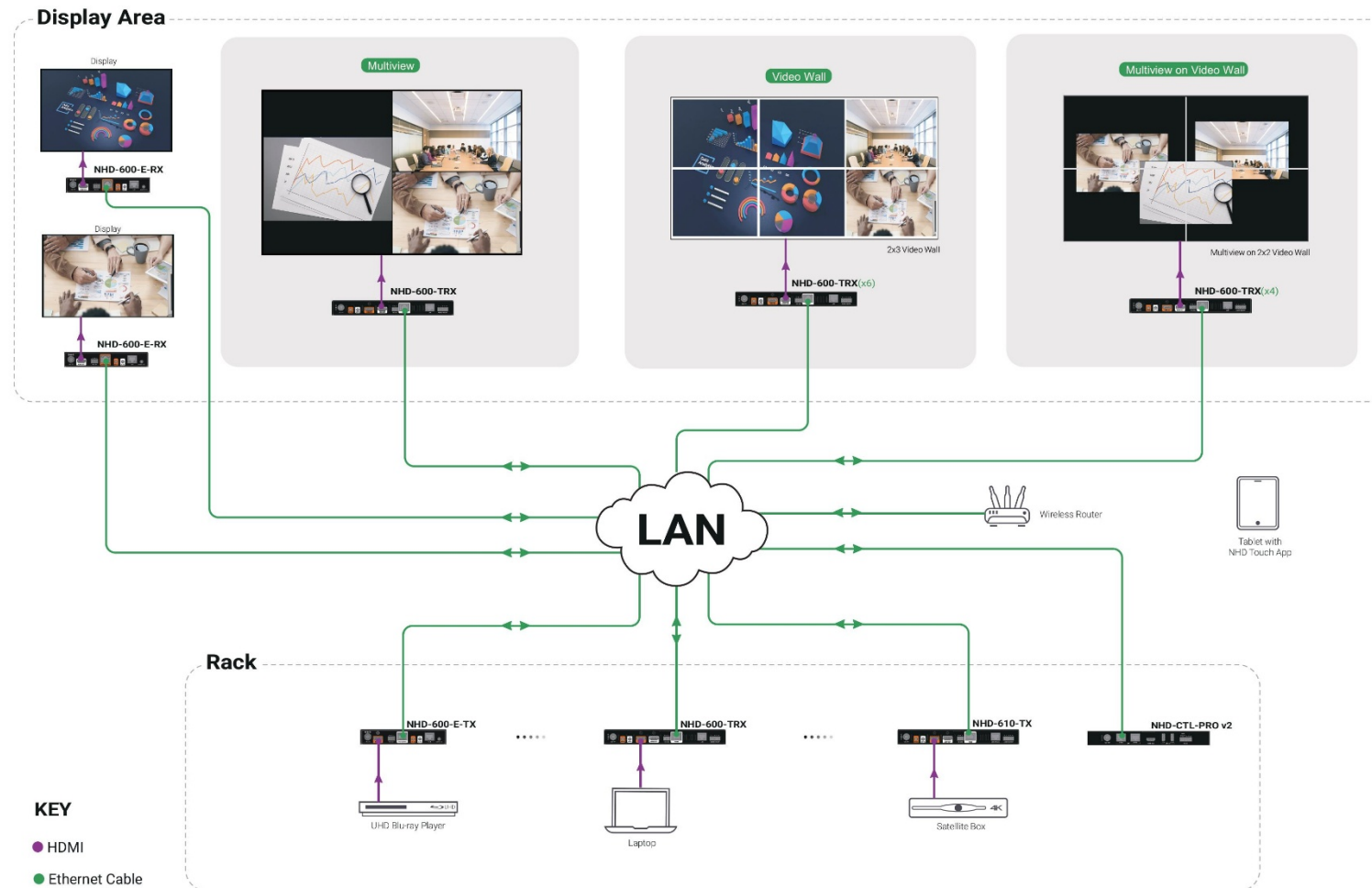
Specification		
Video	Video Interface	1x HDMI 2.0(For TX/RX)
	HDCP	HDCP 2.3 compliant
	Max. Video Resolution	*5120x2160@60Hz 4:2:0, 4096x2160@60Hz 4:4:4, 3840x2160@60Hz 4:4:4 *5K resolution only support in genlock mode
	Scaler	Unsupported
Audio		1x 3.5mm analog
Ethernet	10G Ethernet	1x RJ45
	1G Ethernet	1x RJ45
Control	InfraRed	1x IR In 1x IR Out
	RS232	1x 3-pin Phoenix
Latency		< 120μs

What's in the Box?

- 1x NHD-600-E-TX or NHD-600-E-RX
- 1x IR Transmitter(For TX) or Receiver(For RX)
- 2x Pair of Wall Mounting Ears
- 1x 12V 1A DC PSU with UK/US/EU/AU Pins
- 1x 3-pin 3.5mm Terminal Block

Typical Connection & Applications

The NHD-600-E is designed for point-to-point or point-to-many scenarios. It offers a cost-effective solution for zero-latency applications by removing USB, video wall, multiview and output scaling functions. The typical wiring diagram and application example are shown below.



Comparison with NetworkHD 600 Series

		NHD-600-TRX	NHD-600-TRXF	NHD-610-TX v2	NHD-600-E-TX	NHD-600-E-RX
Device Type		Transceiver	Transceiver	Encoder	Encoder	Decoder
Video	Input	1x HDMI	1x HDMI	1x HDMI	1x HDMI	No
	Output	1x HDMI	1x HDMI	1x HDMI(Loop out)	No	1x HDMI
Audio	Analog Audio	1x IN/OUT	1x IN/OUT	1x IN/OUT	1x IN/OUT	1x OUT
	Dante Audio	No	No	Yes, also AES67	No	No
Control	RS-232	Yes	Yes	Yes	Yes	Yes
	IR	1x IR IN, 1x IR OUT	1x IR IN, 1x IR OUT	1x IR IN, 1x IR OUT	1x IR IN, 1x IR OUT	1x IR IN, 1x IR OUT
	USB	3x USB-A 1.1 1x USB-B 1.1	3x USB-A 2.0 1x USB-B 2.0	3x USB-A 1.1 1x USB-B 1.1	No	No
Other features	SPF+	No	Yes	No	No	No
	RJ45	1x 1G (For control) 1x 10G(For AV)	1x 1G(For control)	1x 1G(For control) 1x 10G(For AV)	1x 1G (For control) 1x 10G(For AV)	1x 1G (For control) 1x 10G(For AV)
	PoE+	Yes	No	Yes	Yes	Yes
	Video Wall	Yes, up to 8x8	Yes, up to 8x8	Yes, up to 8x8	Yes	No
	Multiview	Yes, up to 16 windows	Yes, up to 16 windows	Yes, up to 16 windows	No	No

Comparison with Other Competitors

Specification		WyreStorm NHD-600-E-TX/RX	Lightware HDMI-TPX-TX/RX107
Video	Video Interface	1x HDMI 2.0(For TX/RX)	1x HDMI 2.0(For TX/RX)
	HDCP	HDCP 2.3 compliant	HDCP 2.3 compliant
	Max. Video Resolution	Up to *5120x2160@60Hz 4:2:0, 4096x2160@60Hz 4:4:4, 3840x2160@60Hz 4:4:4	Up to 4096x2160@60Hz 4:4:4 or 3840x2160@60Hz 4:4:4
	Scaler	Unsupported	Unsupported
Audio		1x 3.5mm analog	N/A
Ethernet	10G Ethernet	1x RJ45	1x RJ45
	1G Ethernet	1x RJ45	2x RJ45
Control	InfraRed	1x IR In 1x IR Out	1x IR Out
	RS232	1x 3-pin Phoenix	1x 3-pin Phoenix
Latency		< 120μs	< 120μs

Final Thoughts

Q & A Session

Got questions? Don't hesitate to ask away.

